Inside *zipfR*

Stefan Evert
https://zipfR.R-forge.R-project.org/

typeset on March 11, 2026

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1 The mathematics of LNRE modelling

1.1 Notation

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1.5 Posterior distribution & Good-Turing

2 Productivity & lexical diversity

2.1 Overview of productivity measures

$$\mathrm{TTR} = \frac{V}{N} \tag{1}$$

$$\text{Guiraud's } R = \frac{V}{\sqrt{N}} = \sqrt{N} \cdot \mathrm{TTR} \tag{2}$$

Carroll's $\mathrm{CTTR} = V/\sqrt{2N} = R/\sqrt{2}$ is fully equivalent.

$$\text{Herdan's } C = \frac{\log V}{\log N} = \frac{\log \mathrm{TTR}}{\log N} + 1 \tag{3}$$

Herdan assumes the general power law $V \sim N^{\alpha}$, with $C \to \alpha$ for $N \to \infty$. The assumption is met approximately by any infinite Zipf-Mandelbrot population and $C \to 1/a$.

$$\text{Dugast's } k = \frac{\log V}{\log \log N} = \frac{\log(N \cdot \mathrm{TTR})}{\log \log N} \tag{4}$$

$$\text{Dugast's } U = \frac{(\log N)^2}{\log N - \log V} = \frac{\log N}{1 - C} \tag{5}$$

Maas's $a^2 = (\log N - \log V)/(\log N)^2 = 1/U$ is fully equivalent, but formulated as a measure of "lexical poverty", i.e. low values indicate high productivity.

$$\text{Brunet's } W = N^{V^{-a}} \text{ with } a = 0.172 \tag{6}$$

which looks less ridiculous in the form $\log W = V^{-a} \cdot \log N$

$$\text{Somers's } S = \frac{\log \log V}{\log \log N} \tag{7}$$

is not implemented in *zipfR* because the symbol clashes with Sichel's S and because it's taking the double logs to absurdity.

Baayen (1992) proposes the productivity index

$$\mathscr{P} = \frac{V_1}{N}$$

(originially from his PhD thesis, Baayen 1989), which corresponds to the slope of the vocabulary growth curve.

$$\text{Honoré's } H = 100 \frac{\log N}{1 - V_1/V} \tag{8}$$

$$\text{Sichel's } S = \frac{V_2}{V} \tag{9}$$

Michéa's $M = V/V_2 = 1/S$ is a measure of lexical poverty and fully equivalent to Sichel's S.

For an infinite Zipf-Mandelbrot population, the slope parameter $\alpha = 1/a$ can directly be estimated from the proportions of hapax (Evert 2004: 130) and dis legomena (Evert 2004: 127), which are independent of sample size under certain simplifying assumptions and large-sample approximations:

$$\text{Stefan's } \alpha_1 = \frac{V_1}{V} \quad \text{and} \quad \alpha_2 = 1 - 2\frac{V_2}{V_1}$$

First experiments show that α_1 and α_2 work well for random samples from a ZM population, but can be very sensitive to deviations, esp. in the form of a finite population. Of particular interest is the measure α_1, which is simply the proportion of hapaxes (in analogy to S), and which has also been suggested as an estimator for the Zipf slope parameter by Rouault (1978: 172).

Also note that Sichel's S can be seen as a combination of the two measures:

$$S = \alpha_1 \frac{1 - \alpha_2}{2}$$

Entropy H needs to be distinguished from Honoré's H:

$$H = -\sum_{i=1}^{\infty} \frac{f_i}{N} \log_2 \frac{f_i}{N} = -\sum_{m=1}^{\infty} V_m \frac{m}{N} \log_2 \frac{m}{N} \tag{10}$$

Maximum value depends on V, viz. $H \leq \log_2 V$. Hence compute normalized entropy (aka evenness or efficiency)

$$\eta = \frac{H}{\log_2 V} \tag{11}$$

with $0 \leq \eta \leq 1$. (Or better η^{-1} as productivity measure?) η may be problematic due to counter-intuitive scaling with sample size, and a quick Web search suggests that unbiased estimation of H is really difficult, so we will not pursue H as a productivity measure (but may implement it do demonstrate problematic issues).

Yule suggested a measure based on statistical moments of the frequency spectrum (which sounds quite absurd but could possibly be motivated in terms of sampling random types), leading to

$$\text{Yule's } K = 10^4 \left(-\frac{1}{N} + \sum_{m=1}^{\infty} V_m \left(\frac{m}{N}\right)^2 \right) = 10^4 \cdot \frac{\sum_{m=1}^{\infty} m^2 \cdot V_m - N}{N^2} = 10^4 \cdot \sum_{i=1}^{\infty} \frac{f_i(f_i - 1)}{N^2} \tag{12}$$

Herdan proposed a very similar measure $v_m \approx \sqrt{K}$ based on a different mathematical derviation:

$$\text{Herdan's } v_m = \sqrt{-\frac{1}{V} + \sum_{i=1}^{\infty} V_i \left(\frac{i}{N}\right)^2} \tag{13}$$

Simpson proposed a closely related measure that can be interpreted as an unbiased estimator of a population coefficient $\delta = \sum_i \pi_i^2$, i.e. the probability of drawing the same type twice from the population.

$$\text{Simpson's } D = \sum_{m=1}^{\infty} V_m \frac{m}{N} \frac{m-1}{N-1} = \sum_{i=1}^{\infty} \frac{f_i}{N} \frac{f_i - 1}{N-1} \tag{14}$$

For a LNRE population, the coefficient δ corresponds to the second moment of type density function:

$$\delta = \int_0^{\infty} \pi^2 g(\pi)\, d\pi \tag{15}$$

Baayen *et al.* (1996: 124) already note that "[t]he values of both D and K are primarily determined by the high end of the frequency distribution structure".

2.2 Expected values and sampling distribution

Linear transformations of V or V_m Expectations and full sampling distributions can directly be obtained for TTR and other measures that are linear transformations of V or a single spectrum element V_m. In particular:

$$E\big[\text{TTR}\big] = \frac{E\big[V\big]}{N} \tag{16}$$

$$E\big[R\big] = \frac{E\big[V\big]}{\sqrt{N}} \tag{17}$$

$$E\big[\mathscr{P}\big] = \frac{E\big[V_1\big]}{N} \tag{18}$$

Nonlinear transformations of V For a nonlinear transformation $f(V)$, approximate expectations can be obtained if the distribution of V covers a region in which f is approximately linear. In this case,

$$E\big[f(V)\big] \approx f(E\big[V\big]) \tag{19}$$

As long as f is monotonic, the corresponding equality for the median is always exact. Provided that V has a symmetric distribution (e.g. by being approximately normal), we obtain

$$\text{med}\big[f(V)\big] = f(\text{med}\big[V\big]) \approx f(E\big[V\big]) \tag{20}$$

We assume that these conditions are met for all productivity measures of this form, but may want to check later whether linearity of the transformation is always a plausible assumption.

$$E\big[C\big] = \frac{\log E\big[V\big]}{\log N} \tag{21}$$

$$E\big[k\big] = \frac{\log E\big[V\big]}{\log \log N} \tag{22}$$

$$E\big[U\big] = \frac{(\log N)^2}{\log N - \log E\big[V\big]} \tag{23}$$

$$E\big[\log W\big] = E\big[V\big]^{-a} \cdot \log N \text{ with } a = 0.172 \tag{24}$$

W seems to have (only) slightly larger curvature than $\log W$, so both forms are equally viable.

Measures involving ratios of V and V_m For such measures, approximate expectations can be obtained from normal approximations to V and V_m (which should be very good in the range of samples where the measures are reasonable) and the fact that the ratio of two independet normal variables $X \sim N(\mu_1, \sigma_1^2), Y \sim N(\mu_2, \sigma_2^2)$ can itself be approximated by a normal distribution

$$\frac{X}{Y} \sim N(\mu, \sigma^2) \quad \text{with} \quad \mu = \frac{\mu_1}{\mu_2}, \;\; \sigma^2 = \mu^2 \left(\frac{\sigma_1^2}{\mu_1^2} + \frac{\sigma_2^2}{\mu_2^2} \right) \tag{25}$$

provided that $\mu_i/\sigma_i \gg 1$ (Díaz-Francés and Rubio 2013: 313). This should also hold for correlated variables X and Y with the same μ but different σ^2, as demonstrated for the special case $X/(X+Y)$ by Evert (2004: Lemma A.8). For nonlinear transformations of such ratios, we assume

$$E\left[f\left(\frac{X}{Y} \right) \right] \approx f\left(E\left[\frac{X}{Y} \right] \right) \approx f\left(\frac{E\big[X\big]}{E\big[Y\big]} \right) \tag{26}$$

This yields expectations for:

$$E\big[S\big] = \frac{E\big[V_2\big]}{E\big[V\big]} \tag{27}$$

$$E\big[\alpha_1\big] = \frac{E\big[V_1\big]}{E\big[V\big]} \tag{28}$$

$$E\big[\alpha_2\big] = 1 - 2\frac{E\big[V_2\big]}{E\big[V_1\big]} \tag{29}$$

$$E\big[H\big] = 100\frac{\log N}{1 - E\big[V_1\big]/E\big[V\big]} \tag{30}$$

Variance and sampling distribution Except in the case of linear transformations of V or V_m, the full sampling distributions are much harder to compute. Approximate variances could be determined from normal approximations to V and V_m together with (25) and a linear approximation

$$f(X) \approx f(E\big[X\big]) + f'(E\big[X\big]) \cdot (X - E\big[X\big])$$

Variances would have to be worked out in detail and require a version of (25) for correlated variables, possibly based on the proof by Evert (2004: Lemma A.8).

For the time being, variances and confidence intervals will be determined empirically by parametric boostrapping.

Measures based on the full frequency spectrum Expectations for Simpson D and Yule K are derived from the individual binomial distributions of $f_i \sim B(N, \pi_i)$; additivity of expected values does not required independence of the random variables. Since $E\big[f_i\big] = N\pi_i$ and $\mathrm{Var}\big[f_i\big] = N\pi_i(1-\pi_i)$, we find that

$$E\big[f_i^2\big] = \mathrm{Var}\big[f_i^2\big] + E\big[f_i\big]^2 = N\pi_i(1-\pi_i) + (N\pi_i)^2 \tag{31}$$

Application to Simpson's D yields

$$\begin{aligned} E\big[D\big] &= \frac{1}{N(N-1)} \sum_{i=1}^{\infty} \big(E\big[f_i^2\big] - E\big[f_i\big] \big) \\ &= \frac{1}{N(N-1)} \sum_{i=1}^{\infty} \big(N\pi_i(1-\pi_i) + (N\pi_i)^2 - N\pi_i \big) \\ &= \frac{1}{N(N-1)} \sum_{i=1}^{\infty} \pi_i^2 (N^2 - N) = \sum_{i=1}^{\infty} \pi_i^2 = \delta \end{aligned}$$

proving the claim that D is an unbiased estimator of the population coefficient δ (Simpson 1949: 688).

If we approximate the binomial distributions of f_i with Poisson distributions (as in the simplified Poisson sampling approach to LNRE models), we have $E\big[f_i\big] = N\pi_i$ and $E\big[f_i^2\big] = N\pi_i + (N\pi_i)^2$. Under these assumptions, Yule's K becomes an unbiased estimator:

$$\begin{aligned} E\big[K\big] &= \frac{10^4}{N^2} \cdot \sum_{i=1}^{\infty} \big(E\big[f_i^2\big] - E\big[f_i\big]\big) \\ &= \frac{10^4}{N^2} \cdot \sum_{i=1}^{\infty} N^2 \pi_i^2 = 10^4 \cdot \sum_{i=1}^{\infty} \pi_i^2 = 10^4 \cdot \delta \end{aligned}$$

With binomial sampling, the expectation is $E\big[K\big] = 10^4 \frac{N-1}{N} \delta$.

An equation for the variance of D is provided by Simpson (1949: 688) and may be accepted without further proof, given that his claim about $E\big[D\big]$ was correct.

3 Specific LNRE models

3.1 Zipf-Mandelbrot (ZM)

The second moment δ of the tdf (15) is easily obtained from (33) by setting $A = 0$ (which also applies to the normalizing constant C):

$$\delta = \frac{C}{2-\alpha} B^{2-\alpha} = \frac{(1-\alpha)B^{2-\alpha}}{(2-\alpha)B^{1-\alpha}} \tag{32}$$

3.2 Finite Zipf-Mandelbrot (fZM)

The second moment δ of the tdf (15) is given by

$$\begin{aligned} \delta &= \int_0^{\infty} \pi^2 g(\pi)\, d\pi = C \int_A^B \pi^{1-\alpha}\, d\pi \\ &= C \left[\frac{\pi^{2-\alpha}}{2-\alpha}\right]_A^B = \frac{C}{2-\alpha} \left(B^{2-\alpha} - A^{2-\alpha}\right) \\ &= \frac{1-\alpha}{2-\alpha} \cdot \frac{B^{2-\alpha} - A^{2-\alpha}}{B^{1-\alpha} - A^{1-\alpha}} \end{aligned} \tag{33}$$

3.3 Generalised Zipf-Mandelbrot (gZM)

3.4 Generalised Inverse Gauss-Poisson (GIGP)

3.5 Montemurro / Tsallis

Based on original research by Tsallis, Montemurro (2001) proposes the following type density function: Insert references to Tsallis

$$g(\pi) = C \left(\mu\pi^R + (\lambda - \mu)\pi^Q\right)^{-1} \tag{34}$$

with parameters $1 < R < Q$ and $\mu, \lambda \in \mathbb{R}$, normalising constant C and possible restriction to a suitable region $A \leq \pi \leq B$. Eq. (34) is derived from a differential equation for the rank-frequency relationship, which does not have a closed-form rank-frequency solution in the general case. See Sec. 5.1.1 for motivation and details.

It will be difficult to obtain closed-form solutions for $E\big[V\big], E\big[V_m\big]$ and other relevant quantities from Eq. (34), and numerical integration may often be required.

Montemurro's LNRE model can be thought of as a smooth interpolation between two power laws with different slopes that hold in different frequency ranges. Therefore, a two-segment gZM model (Sec. 3.3) or a mixture of two ZM/gZM models (Sec. 3.6) should give an approximation to Eq. (34), but will be much easier to handle mathematically (closed-form solutions, numerical accuracy).

3.6 Mixture models

A mixture model describes the LNRE population as a mixture of $K \geq 2$ **component** models $M^{(k)}$ with weights $w^{(k)}$ such that $\sum_{k=1}^{K} w^{(k)} = 1$. The individual $M^{(k)}$ are usually standard LNRE distributions like ZM or fZM. Mixtures can be defined in three different ways:

1. A weighted sum of type density functions, so that

$$g(\pi) = w_1 g_1(\pi) + \ldots + w_K g_K(\pi) \tag{35}$$

   This approach corresponds most closely to standard statistical mixture distributions.

2. Each component $M^{(k)}$ represents a distinct set of types that account for a total proportion w_k of the probability mass (and a corresponding percentage of the tokens in a sample drawn from the population). This approach is intuitively appealing and plausible for real-life data: components might represent the different distributions of nouns, verbs, adjectives, ... or the different subpatterns of a morphological suffix (cf. Baayen 2001: Sec. 4.3). Since each component accounts for a part of the tokens only, type probabilities have to be shifted accordingly in the mixture density:

$$g(\pi) = g_1(w_1 \pi) + \ldots + g_K(w_K \pi) \tag{36}$$

3. All components cover the same set of type, representing their distributions in different parts of a sample. While this is a realistic situation, e.g. components for written and spoken language in a reference corpus, it is mathematically complicated. Since the types will usually not be ranked in the same order in the different components, the mixture distribution can only be derived from a K-dimensional joint type density of the form:

$$g(\pi^{(1)}, \ldots, \pi^{(K)}) \tag{37}$$

Mixture types 1 and 2 are fully equivalent and lead to a mixture type density in the form of a weighted sum. They differ only in how the individual component densities are re-scaled in the combination.[1] Mixture type 3 is impractical, since the multidimensional joint type density can only be estimated from very large samples and may require entirely different strategies than standard LNRE models.

Following Baayen (2001: Sec. 4), we adopt approach 2, which has a more intuitive interpretation than approach 1: each component describes the "internal" probability distribution of a subset of the types, and random samples of size N can be generated by sampling $\approx w_k N$ tokens from $M^{(k)}$.

Spell out mathematics of type 2 mixture models in detail, then experimental implementation in *zipfR*.

- r.v. $f_i^{(k)}$ for type frequencies in subsamples from the different components are distinct and mutually independent (because the subsamples are independent)

[1] Type 1: thinning out types in theoriginal probability range; type 2: keeping the same number of types but down-scaling their probabilities.

- hence the partial spectrum elements $\mathring{V}_m^{(k)}$ and vocabulary sizes $\mathring{V}^{(k)}$ are also independent, so we have simple additivity for expectations, variances and covariances, e.g.

$$E\big[V\big] = \sum_{k=1}^{K} E\big[\mathring{V}^{(k)}\big] \quad \text{and} \quad \mathrm{Var}\big[V\big] = \sum_{k=1}^{K} \mathrm{Var}\big[\mathring{V}^{(k)}\big] \tag{38}$$

- component $M^{(k)}$ accounts for an expected number $w_k N$ of tokens in a sample of size N, so expectation and variances for the partial vocabulary size and spectrum elements are

$$E\big[\mathring{V}^{(k)}\big] = E\big[V^{(k)}(w_k N)\big] \qquad E\big[\mathring{V}_m^{(k)}\big] = E\big[V_m^{(k)}(w_k N)\big] \tag{39}$$

$$\mathrm{Var}\big[\mathring{V}^{(k)}\big] = \mathrm{Var}\big[V^{(k)}(w_k N)\big] \qquad \mathrm{Var}\big[\mathring{V}_m^{(k)}\big] = \mathrm{Var}\big[V_m^{(k)}(w_k N)\big] \tag{40}$$

- confidence/prediction intervals, goodness-of-fit tests, etc. can be computed based on these expectations, variances and covariances

- type and probability densities, cumulative type and probability distribution, quantiles and the posterior distribution can be determined from the mixture type density (36); quantiles, and MAP will often be a little tricky (because they have to take the overlapping component densities into account); $E\big[V_{m,\rho}\big]$ and $E\big[V_{m,>\rho}\big]$ for given ρ are additive, though

- population diversity is additive because the type mass of each component is not rescaled in the mixture, i.e.

$$S = S^{(1)} + \ldots + S^{(K)} \tag{41}$$

- generating random samples is particularly easy for type-2 mixtures; for a sample of N tokens, draw component sizes from a multinomial distribution

$$(N^{(1)}, \ldots, N^{(K)}) \sim \mathrm{Mult}(N, w_1, \ldots, w_K) \tag{42}$$

  then sample $N^{(k)}$ tokens from component $M^{(k)}$ and aggregate the results

- parameter estimation for mixture models can in principle follow the same approach as for standard LNRE models, especially if component weights are pre-determined (otherwise they have to be constrained to add up to 1); however, the much larger number of parameters will require a large sample and inclusion of more spectrum elements; Baayen (2001: Ch. 4) doesn't address the issue at all

- in many cases, it might be better to pre-split the training data, e.g. by part of speech or into non-overlapping frequency bands, and estimate component models separately; this approach could even be used to capture lexicalised types with a fZM with very small population diversity S

4 Applications and extensions

4.1 Vocabulary overlap

- one important application is the comparison of two samples A and B from populations containing the same types, though possibly with different frequency distributions; for convenience, we also refer to the populations as A and B

- formally, we assume a joint vocabulary $\{1, \ldots, S\}$ for the two populations; type i has occurrence probability π_i in A and π'_i in B, each of which may be zero (if a type doesn't occur in this population); types are listed by descending sum of the probabilities $\pi_i + \pi'_i$, which may be different from the canonical ordering for either population; importantly, i always refers to *the same* type in both populations

- type frequency counts are given by r.v. f_i for A and f'_i for B; the individual frequency spectra of the samples are denoted by V, V_m for A and V', V'_m for B; and the sample sizes are N, N'
- the key statistic for analysing vocabulary overlap is the **joint frequency spectrum** (jfs)

$$V_{k,k'} = \sum_{i=1}^{S} I_{\left[f_i = k \wedge f'_i = k'\right]}$$

which takes the form of a two-dimensional table

	V'_0	V'_1	V'_2	V'_3	V'_4	V'_5	$\dots$
V_0	$V_{0,0}$	$V_{0,1}$	$V_{0,2}$	$V_{0,3}$	$V_{0,4}$	$V_{0,5}$	$\dots$
V_1	$V_{1,0}$	$V_{1,1}$	$V_{1,2}$	$V_{1,3}$	$V_{1,4}$	$V_{1,5}$	$\dots$
V_2	$V_{2,0}$	$V_{2,1}$	$V_{2,2}$	$V_{2,3}$	$V_{2,4}$	$V_{2,5}$	$\dots$
V_3	$V_{3,0}$	$V_{3,1}$	$V_{3,2}$	$V_{3,3}$	$V_{3,4}$	$V_{3,5}$	$\dots$
V_4	$V_{4,0}$	$V_{4,1}$	$V_{4,2}$	$V_{4,3}$	$V_{4,4}$	$V_{4,5}$	$\dots$
V_5	$V_{5,0}$	$V_{5,1}$	$V_{5,2}$	$V_{5,3}$	$V_{5,4}$	$V_{5,5}$	$\dots$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\ddots$

- the jfs is usually much sparser than the spectra of individual samples, which makes the usual approach based on normal approximations unsuitable, except for the very low end of the jfs and/or very large samples
- the marginal spectra are obtained by summing over the rows and columns of the jfs:

$$V_m = \sum_{k'} V_{m,k'} \qquad V'_{m'} = \sum_{k} V_{k,m'}$$

and the marginal vocabulary sizes by summing over everything except for the first row or first column, respectively

$$V = \sum_{k=1}^{\infty} \sum_{k'} V_{k,k'} \qquad V' = \sum_{k} \sum_{k'=1}^{\infty} V_{k,k'}$$

- the vocabulary size and frequency spectrum of the union $A \cup B$ of both samples are obtained by summing over diagonals:

$$V^{A \cup B} = \sum_{k+k'>0} V_{k,k'} = S - V_{0,0}$$

$$V_m^{A \cup B} = \sum_{k'=0}^{m} V_{m-k',k'}$$

- the overlap vocabulary size $V^{A \cap B}$ is obtained by summing over everything except for the first row and column

$$V^{A \cap B} = \sum_{k,k'=1}^{\infty} V_{k,k'}$$

and can also be broken down by total frequency $m = f_i + f'_i$ of each shared type, which requires summing over shortened diagonals (excluding $V_{m,0}$ and $V_{0,m}$)

$$V_m^{A \cap B} = \sum_{k'=1}^{m-1} V_{m-k',k'}$$

- the number of types unique to A is the sum over the first column (except for $V_{0,0}$), and analogously for the number of types unique to B

$$V^{A \setminus B} = \sum_{k=1}^{\infty} V_{k,0} \qquad V^{B \setminus A} = \sum_{k'=1}^{\infty} V_{0,k'}$$

- this results in several simple identities

$$
\begin{aligned}
V^{A\cup B} &= V + V^{A\setminus B} = V' + V^{B\setminus A} \\
V^{A\cap B} &= V - V^{A\setminus B} = V' - V^{B\setminus A} = V + V' - V^{A\cup B} \\
V_m^{A\cap B} &= V_m^{A\cup B} - V_{0,m} - V_{m,0}
\end{aligned}
$$

- finally, the shared token count of A and B, i.e. the total number of tokens for the shared types, can be computed as

$$
N^{A\cap B} = \sum_{m=1}^{\infty} m V_m^{A\cap B}
$$

4.1.1 Two samples from the same population

- the only relatively easy case is when A an B are two independent samples from the same population
- as usual, we assume independent Poisson sampling in order to simplify equations and to have a better chance of being able to work out some variances at least
- the Poisson probability of type i occurring k times in a sample of size N is

$$
E\big[I_{[f_i=k]}\big] = \Pr\big(f_i = k\big) = e^{-N\pi_i}\frac{(N\pi_i)^k}{k!}
$$

- since A and B are independent samples, the joint probability is

$$
\begin{aligned}
E\big[I_{[f_i=k\wedge f_i'=k']}\big] &= \Pr\big(f_i = k \wedge f_i' = k'\big) \\
&= e^{-N\pi_i}\frac{(N\pi_i)^k}{k!} \cdot e^{-N'\pi_i}\frac{(N'\pi_i)^{k'}}{k'!} \\
&= e^{-(N+N')\pi_i}\frac{N^k (N')^{k'}}{k!k'!}\pi_i^{k+k'} \\
&= e^{-(N+N')\pi_i}\frac{\big((N+N')\pi_i\big)^{k+k'}}{(k+k')!} \cdot \binom{k+k'}{k} \cdot \frac{N^k (N')^{k'}}{(N+N')^{k+k'}} \\
&= e^{-(N+N')\pi_i}\frac{\big((N+N')\pi_i\big)^{k+k'}}{(k+k')!} \cdot B\big(k; k+k', \tfrac{N}{N+N'}\big)
\end{aligned}
$$

  with the binomial probability

$$
B\big(k; n, p\big) = \binom{n}{k} p^k (1-p)^{n-k}
$$

- the intuition behind this result is simple: the spectrum element $V_m^{A\cup B}$ represents types with a total frequency of $m = k + k'$ in the union of the samples; it is the sum over a diagonal in the jfs; types are assigned to the individual jfs cells depending on how their m occurrences are split across the two samples, which leads to a binomial distribution
- for identical sample sizes $N = N'$ the equation simplifies further:

$$
E\big[I_{[f_i=k\wedge f_i'=k']}\big] = e^{-2N\pi_i}\frac{(2N\pi_i)^{k+k'}}{(k+k')!} \cdot B\big(k; k+k', \tfrac{1}{2}\big)
$$

- we can easily obtain expected spectrum sizes by summing over the probabilities

$$E\big[V_m\big] = \sum_{i=1}^{S} E\big[I_{[f_i=m]}\big]$$

and correspondingly for the jfs

$$E\big[V_{k,k'}\big] = \sum_{i=1}^{S} E\big[I_{[f_i=k \wedge f'_i=k']}\big]$$

- for a practical implementation, the sum must be approximated by a Poisson integral over the type density function $g(\pi)$:

$$E\big[V_m\big] = \int_0^\infty \frac{(N\pi)^m}{m!} e^{-N\pi} g(\pi)\, d\pi$$

- after the transformation above, the expectation of the jfs also takes the form of a Poisson integral for the union of the samples (with frequency class $m = k+k'$ and sample size $N+N'$):

$$\begin{aligned} E\big[V_{k,k'}\big] &= \int_0^\infty \frac{\big((N+N')\pi\big)^{k+k'}}{(k+k')!} e^{-(N+N')\pi} \cdot B\big(k; k+k', \tfrac{N}{N+N'}\big) \cdot g(\pi)\, d\pi \\ &= B\big(k; k+k', \tfrac{N}{N+N'}\big) \cdot E\big[V_{k+k'}(N+N')\big] \end{aligned}$$

- expected vocabulary size is computed analogously via the identity $I_{[f_i>0]} = 1 - I_{[f_i=0]}$

$$E\big[V\big] = \sum_{i=1}^{S} E\big[I_{[f_i>0]}\big] = \sum_{i=1}^{S} E\big[1 - I_{[f_i>0]}\big] = \sum_{i=1}^{S} \big(1 - e^{-N\pi_i}\big)$$

and approximated via the integral

$$E\big[V\big] = \int_0^\infty \big(1 - e^{-N\pi}\big) g(\pi)\, d\pi$$

- we can compute expected vocabulary sizes for the union of the samples and their vocabulary differences (types unique to one sample) in a similar way

$$\begin{aligned} E\big[V^{A\cup B}\big] &= \sum_{i=1}^{S} E\big[I_{[f_i>0 \vee f'_i>0]}\big] = \sum_{i=1}^{S} E\big[1 - I_{[f_i=0 \wedge f'_i=0]}\big] = \sum_{i=1}^{S} \big(1 - e^{-N\pi_i} e^{-N'\pi_i}\big) \\ &= \sum_{i=1}^{S} \big(1 - e^{-(N+N')\pi_i}\big) = E\big[V(N+N')\big] \\ E\big[V^{A\setminus B}\big] &= \sum_{i=1}^{S} E\big[I_{[f_i>0 \wedge f'_i=0]}\big] = \sum_{i=1}^{S} \big(1 - e^{-N\pi_i}\big) e^{-N'\pi_i} \\ &= \sum_{i=1}^{S} \big(1 - e^{-(N+N')\pi_i}\big) - \big(1 - e^{-N'\pi_i}\big) = E\big[V(N+N')\big] - E\big[V(N')\big] \\ E\big[V^{B\setminus A}\big] &= E\big[V(N+N')\big] - E\big[V(N)\big] \end{aligned}$$

- based on these results, we obtain the expected vocabulary overlap as

$$\begin{aligned} E\big[V^{A\cap B}(N,N')\big] &= \sum_{i=1}^{S} E\big[I_{[f_i>0 \wedge f'_i>0]}\big] = \sum_{i=1}^{S} \big(1 - e^{-N\pi_i}\big) \cdot \big(1 - e^{-N'\pi_i}\big) \\ &= \sum_{i=1}^{S} \big(1 - e^{-N\pi_i}\big) - \sum_{i=1}^{S} \big(1 - e^{-N\pi_i}\big) e^{-N'\pi_i} = E\big[V(N)\big] - E\big[V^{A\setminus B}\big] \\ &= E\big[V(N)\big] - \big(E\big[V(N+N')\big] - E\big[V(N')\big]\big) \\ &= E\big[V(N)\big] + E\big[V(N')\big] - E\big[V(N+N')\big] \end{aligned}$$

re-derive these results with the easy intuitive reasoning of a joint random sample of size $N = N + N'$

simple equation for expected frequency spectrum of overlap?

explain why variances and the full sampling distributions are difficult and will first have to be explored via simulation experiments

4.1.2 Samples from different populations

- two populations with shared vocabulary are described by pairs of probabilities (π_i, π'_i) for corresponding types, and approximated by a **joint type density function** (jtdf) $g(\pi, \pi')$
- the baseline case of two populations with uncorrelated type probabilties

$$g(\pi, \pi') = g(\pi) \cdot g(\pi')$$

  is extremely unrealistic and might only serve as a purely mathematical point of reference (e.g. *the* might then be the most common word in A but extremely rare in B); as a consequence, any realistic model will have to assume a relatively high level of correlation between the frequency distributions
- the other extreme is the case of two identical populations A and B (or, equivalently, samples from the same population, as discussed in Sec. 4.1.1); in this case, the type mass is concentrated along the diagonal, resulting in some improper delta function similar to

$$g(\pi, \pi') = g(\pi)\delta(\pi' - \pi)$$

- this case therefore needs to be analysed separately (Sec. 4.1.1); however, this observation also means that very strongly correlated populations cannot easily be represented by a jtdf, which would be concentrated on a small area close to the diagonal and hence be poorly conditioned

5 Literature notes

5.1 Extensions of Zipf's law

5.1.1 Montemurro (2001)

Montemurro (2001) proposes an extension to Zipf's law that results in a better empirical fit for higher ranks (i.e., low-frequency data), while Zipf-Mandelbrot only improves the fit for low ranks. Based on all-words Zipf rankings for various literary corpora compiled from Gutenberg e-texts, he observes that the original form of Zipf's law (with $a \approx 1$) only seems to hold for a "middle range" of frequency ranks, $r \approx 100 \ldots 2000/6000$ (depending on corpus size). He further claims that higher ranks follow a similar power law with steeper slope ($a \approx 2 \ldots 3$).

In the following, notation has been adjusted to *zipfR* conventions:

- r = Zipf rank (original: s)
- p_r = relative frequency of r-th type in Zipf ranking (original: $f(s)$)
- Q, R = Zipf slopes (original: q, r)

Noting that the Zipf-Mandelbrot law can be derived from a differential equation

$$\frac{dp}{dr} = -\lambda p^Q \tag{43}$$

(Eq. (3), p. 572), he derives a generalisation from the differential equation

$$\frac{dp}{dr} = -\lambda p^R - (\lambda - \mu) p^Q \tag{44}$$

(Eq. (4), p. 572). There are closed-form solutions for the special cases $R = Q = 1$ (Zipf-Mandelbrot) and $R = 1, Q > 1$ (Zipf's law in middle range, steeper slope for higher ranks), but not for the general case $1 < R < Q$ (p. 573).

Empirically, a good fit is obtained for literary corpora from single authors, using the closed form solution with $R = 1, Q > 1$:

$$p_r = \left(1 - \frac{\lambda}{\mu} + \frac{\lambda}{\mu} e^{(Q-1)\mu r}\right)^{-\frac{1}{Q-1}} \tag{45}$$

(Eq. (6), p. 573). For larger, composite corpora, the general form $1 < R < Q$ seems to be required.

The differential equation (44) and its various closed-form or implicit solutions are attributed to Constantino Tsallis:

Find papers by Tsallis & Denisov

- Tsallis/Bemski/Mendes (1999), *Phys Lett A* **257**, 93
- Tsallis (1988), *J Stat Phys* **52**, 479 — underlying framework of statistical mechanics
- Montemurro says that Tsallis has suggested application to linguistic data in "private communication"
- Denisov (1997), *Phys Lett A* **235**, 447 — relates Zipf-Mandelbrot law to "fractal structure of symbolic sequences with long-range correlations" (p. 572)

Montemurro notes that with some approximations, the general case can be expressed in closed form as a type density function (which he awkwardly refers to as a "probability density"), resulting in the LNRE model:

$$g(\pi) \propto \left(\mu \pi^R + (\lambda - \mu) \pi^Q\right)^{-1} \tag{46}$$

See Sec. 3.5 for more information on this LNRE model.

Montemurro postulates that the two power laws may correspond to "general" and "specialised" vocabulary without further evidence: "This suggests ... the vocabularies can be divided into two parts of distinct nature: one of basic usage ..., and a second part containing more specific words with a less flexible syntactic function." (p. 571, citing then unpublished work by Ferrer & Solé). If we accept this claim, a linear mixture model (Sec. 3.6) would be much more appropriate than a hard split at rank $r \approx 2000 \ldots 6000$.

There is a completely unfounded claim at the end of the paper that "it seems quite plausible that there may be a deep connection between differential equation (4) and the actual processes underlying the generation of syntactic language." (p. 577).

6 Notes and ideas

6.1 Mathematics and implementation

- LNRE model fit may be affected by a small number of very high-frequency types, esp. "echo" tokens (Baroni and Evert 2007) or the "other" type when modelling vocabulary growth wrt. all tokens (*external productivity*). It would probably be useful to separate the most

frequent types, estimate their occurrence probabilities directly (MLE are reliable barring non-randomness effects), and apply the LNRE model only to the remaining vocabulary. This should present no major mathematical obstacles, but will have to be taken into account throughout the implementation (expectations, variances, chi-squared statistics, etc.).

- Standard LNRE fitting uses only the low end of the frequency spectrum and may produce an unsatisfactory fit for the "middle range" of the Zipf ranking. If we want to account for these data as well – esp. in connection with mixture models (Sec. 3.6) and gZM (Sec. 3.3) – a different goodness-of-fit goal function will be needed for parameter estimation.

  One possibility is to pool multiple frequency classes together, e.g. on a logarithmic scale: $m = 1, \ldots, 10$, $m = 11 \ldots 14$, $m = 15 \ldots 20$, $m = 21 \ldots 50$, $m = 51 \ldots 100$, $m = 101 \ldots 1000$, etc.; granularity will have to be adjusted to the available data, of course. Assuming the usual multivariate Gaussian joint distriubtion for the original frequency spectrum, the pooled frequency spectrum should also be multivariate Gaussian as a linear map of the original spectrum. Expectations, variances and covariances should be straightforward, although care has to be taken to avoid performance and/or numerical accuracy issues.

Are there simplified equations for expectations and (co)variances of a pooled frequency spectrum?

- Sometimes it would be useful to fit a LNRE model to multiple frequency spectra. E.g. for Gordon Pipa's neural spiking data, where it is plausible that trials for the same condition follow the same Zipfian distribution, but data cannot be pooled directly; or to avoid overfitting of non-random data by parallel parameter estimation from frequency spectra at different sample sizes. Such *co-estimation* should be relatively straightforward to implement by adding up cost functions (perhaps with suitable scaling to account for different sample sizes), but custom estimators available for some of the models can no longer be used.

Is "co-estimation" an appropriate term?

- One problem of the fZM implementation may be numerical accuracy due to cancellation when "short" Gamma integrals are calculated as differences between incomplete Gamma functions, esp. on very small or otherwise extreme samples. This will become much more virulent for gZM models with many components. Suggest a two-step strategy:

  1. Encapsulate finite Gamma integrals into a helper function, which estimates cancellation errors and collects statistics. This should be controlled by a global `debug` option for *zipfR* (set with `zipfR.par()`).
  2. Implement more accurate algorithm for finite Gamma integrals. So far, the only solution seems to be numeric integration, which is easy and accurate for monotonic functions (possibly splitting integrand into monotonic parts). Code might be implemented in R (using standard numeric integration functions) for preliminary testing.
  3. Reimplement numeric integration in C; for better efficiency and accuracy on "long" Gamma integrals, might compute incomplete Gamma function first and run numeric integration only when estimated cancellation error exceeds a pre-defined threshold (also set with `zipfR.par()`).

6.2 Thoughts on goals and applications of LNRE modelling

- Most research on Zipf's law (both Zipf himself and more recent work by physicists) focuses on middle-range frequency ranks, which are highlighted in a logarithmic rank-frequency graph. By contrast, LNRE models (Khmaladze 1987; Baayen 2001) based on truncated frequency spectra are only interested in the lowest-frequency types. Note that for typical applications – productivity, vocabulary growth, estimation of vocabulary diversity, adjusted significance tests – only such lowest-frequency types are of major concern, as probability estimates for middle- and high-frequency data can be obtained directly from any sizable corpus. This is explains why most LNRE models find Zipf slows $a \gg 1$ rather than $a \approx 1$ as observed by Zipf and related work.

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Todo list

- Insert references to Tsallis 6
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